

MID-OHIO VALLEY, WV, USA

WMO: 724273

Lat: 39.345N

Lon: 81.439W

Elev: 831

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03804

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	12.0	-3.1	4.8	10.0	1.8	6.2	15.6	20.9	37.3	19.0	36.1	5.0	260	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	90.5	73.7	88.1	72.7	86.0	71.6	76.7	86.0	75.2	83.8	74.0	82.2	7.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	130.0	81.0	72.5	124.4	79.6	71.5	119.9	78.5	40.5	86.4	39.1	84.1	38.0	81.9	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	16.0	13.6	-1.2	93.7	7.1	3.2	-6.3	96.0	-10.5	97.8	-14.5	99.6	-19.7	101.9
			-2.0	79.5	6.9	1.9	-6.9	80.8	-10.9	81.9	-14.8	83.0	-19.8	84.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	31.9	35.3	43.3	55.0	63.4	70.9	74.3	73.3	67.0	55.4	44.6	36.5
	DBStd	17.36	12.61	11.33	10.84	9.41	7.87	5.62	4.68	4.76	6.78	8.70	9.47	10.24
	HDD50	2023	577	427	266	55	4	0	0	0	0	43	217	434
	HDD65	4871	1025	832	675	320	129	16	1	3	56	317	614	883
	CDD50	3609	17	15	60	204	419	627	753	722	511	211	54	16
	CDD65	980	0	0	3	19	78	193	289	260	117	20	1	0
	CDH74	8412	0	2	32	235	750	1641	2398	2129	1033	185	7	0
	CDH80	2694	0	0	4	39	192	517	843	723	339	37	0	0
Wind	WSAvg	5.5	7.0	7.0	6.9	6.9	5.2	4.6	4.1	3.8	4.1	4.8	5.7	6.2
Precipitation	PrecAvg	41.5	2.6	2.5	4.1	3.4	4.0	3.8	4.3	4.2	3.2	2.9	3.3	3.2
	PrecMax	56.1	5.9	5.4	9.4	7.6	10.5	7.4	8.0	8.3	6.6	7.9	7.9	8.1
	PrecMin	27.2	0.7	0.6	1.0	0.8	0.9	0.4	1.3	0.5	0.6	0.5	0.5	0.4
	PrecStd	6.9	1.3	1.3	2.0	1.7	2.2	1.8	1.9	1.9	1.5	1.8	1.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.5	70.0	78.7	84.3	87.7	91.5	93.3	93.2	92.1	84.2	74.2	67.0
		MCWB	58.3	56.6	60.3	63.8	69.8	73.1	74.7	74.7	71.1	68.9	61.4	58.6
	2%	DB	61.5	63.3	72.5	80.3	84.8	88.2	90.2	89.9	87.8	79.0	69.4	61.9
		MCWB	54.8	53.0	57.0	61.1	68.5	72.9	75.1	73.7	70.2	65.3	57.8	54.4
	5%	DB	56.4	58.2	67.0	76.4	82.0	85.7	87.6	87.0	83.9	74.8	64.8	57.1
		MCWB	51.8	50.5	53.4	59.5	67.5	71.8	73.6	72.6	69.2	62.7	55.0	51.1
	10%	DB	51.1	53.2	61.4	72.1	78.7	83.0	84.8	84.2	80.3	70.7	60.8	52.9
		MCWB	45.5	46.6	51.9	57.3	66.1	70.5	72.4	71.4	67.4	61.1	52.9	47.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	59.7	62.2	67.1	73.4	77.0	79.0	78.3	74.9	71.0	63.8	59.7
		MCDB	65.1	66.5	74.2	79.9	83.7	86.8	88.9	87.3	84.2	80.5	71.0	65.1
	2%	WB	56.0	55.6	59.3	64.2	71.2	75.0	77.1	76.4	72.9	68.1	59.7	56.4
		MCDB	60.0	60.2	68.7	74.6	81.1	84.4	86.6	85.4	82.4	75.7	66.4	60.4
	5%	WB	52.3	51.5	56.6	61.9	69.3	73.5	75.5	74.9	71.2	65.2	56.8	52.3
		MCDB	56.7	57.2	64.0	71.6	78.4	82.6	83.7	83.3	79.6	71.9	62.7	56.1
	10%	WB	46.1	46.5	53.0	59.8	67.6	72.2	74.3	73.4	69.5	62.3	53.7	47.8
		MCDB	50.0	52.3	60.5	69.0	75.9	80.6	81.8	80.9	76.4	69.0	60.3	52.3
Mean Daily Temperature Range	5% DB	MDBR	15.1	17.1	19.8	23.1	21.5	20.2	19.5	20.3	21.6	21.2	18.8	15.0
		MCDBR	18.9	25.0	27.9	28.4	25.7	23.4	22.3	23.2	26.2	26.2	25.0	21.5
		MCWBR	14.3	18.1	15.9	14.4	12.1	10.6	9.4	9.4	10.8	13.6	15.8	16.1
	5% WB	MCDBR	17.5	22.3	22.6	22.6	21.1	19.9	18.8	19.2	21.0	21.7	20.1	18.8
	MCWBR	14.3	18.3	14.8	13.2	10.7	9.8	8.9	9.0	9.6	12.2	15.0	15.8	
Clear-Sky Solar Irradiance	taub	0.307	0.321	0.346	0.389	0.434	0.462	0.483	0.484	0.407	0.364	0.324	0.299	
	taud	2.473	2.421	2.414	2.314	2.198	2.160	2.119	2.092	2.297	2.419	2.506	2.532	
	Ebn at Noon	275	286	289	281	270	261	254	250	265	266	266	270	
	Edh at Noon	25	30	34	40	46	48	49	49	38	30	24	22	
All-Sky Solar Radiation	RadAvg	536	770	1118	1492	1720	1893	1811	1697	1402	976	678	460	
	RadStd	56	113	91	135	143	133	110	101	136	109	90	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	N/A
(47)	Regional (32 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+124

Nomenclature: See separate page